No Keller Map Admits Two Dicritical Divisors:

Closure of the Two-Dicritical Line for the Planar Jacobian Conjecture

Constant Leading Forms, the E_1 Degree Obstruction, and the $m = 1$ Bridge Induction

Brandon D. McCrary

Independent Researcher

ORCID: 0009-0008-2804-7165

Computation and LaTeX Source:

https://github.com/brandonmccraryresearch-cloud/2D-Jacobian-Conjecture-Workspace

October 1, 2026

Abstract

We close the two-dicritical line for the planar Jacobian conjecture: no polynomial map $(P, Q) : \mathbb{C}^2 \to \mathbb{C}^2$ with $[P, Q] \in \mathbb{C}^*$ (a Keller map) admits two distinct dicritical divisors in its resolution. The argument proceeds in two parts.

First, no dicritical of order $m \geq 2$ exists for any Keller map. For every exceptional divisor E_k with $k \geq 2$, a constant-leading-form theorem forces the pencil restriction to have degree 0, ruling out $m \geq 2$. On the first exceptional divisor E_1, local geometry permits a solitary $m = 2$ dicritical, but every monomial allowed by the order condition has total degree at least 2, so the Jacobian has minimum degree 2 and cannot be a nonzero constant.

Second, the $m = 1$ two-branch bridge is empty for every degree. For distinct centers $a_3 \neq a_4$ and arbitrary degree $d \geq 2$, the simultaneous order conditions force every coefficient with positive x-power to vanish. The proof is a descending induction on the weighted degree $W = 2J + K$: each diagonal block is $\left[\binom{e_k}{i} a^{e_k - i}\right]$, which reduces by row operations to a scaled Vandermonde matrix in the distinct e_k, hence has full column rank. The nullspace is exactly $\operatorname{span}\{1, y\}$, so the Jacobian vanishes identically.

Every computational claim is verified by an actually-executed script archived with this paper; §5 maps each script to the exact arithmetic it certifies.

---

1 Introduction

1.1 Context

The planar Jacobian conjecture asserts that a polynomial map $(P, Q) : \mathbb{C}^2 \to \mathbb{C}^2$ with $[P, Q] := P_x Q_y - P_y Q_x \in \mathbb{C}^*$ is invertible. A *Keller map* is such a pair with constant nonzero Jacobian.

A central question in the resolution theory of Keller maps is the structure of dicritical divisors: exceptional divisors on which the pulled-back pencil has no base points. The *two-dicritical line* asks whether a Keller map can admit two distinct dicritical divisors in its resolution. This paper closes that line completely.

1.2 Main result

Theorem 1.1 (No two dicriticals)**.** *Let K be a field of characteristic 0. Let $P, Q \in K[x, y]$ with $[P, Q] = c \in K^\times$. Then the resolution of the pencil $\lambda P + \mu Q$ contains no two distinct dicritical divisors.*

The proof has two independent parts:

1. **Order $m \geq 2$ is impossible** (§2): the constant-leading-form theorem closes all E_k ($k \geq 2$); the E_1 degree obstruction closes the first exceptional divisor.
2. **The $m = 1$ bridge is empty** (§3): a triangular induction on weighted degree shows the two-center $m = 1$ system has nullspace span$\{1, y\}$ for every d.

1.3 Logical status

Remark 1.2. Theorem 1.1 is unconditional: it does not depend on GGHV [1] or any unproved classification. The $m \geq 2$ argument uses only the local order computation and elementary degree counting. The $m = 1$ argument uses only linear algebra (Vandermonde determinants) and the binomial theorem. Every computational verification was executed; scripts and outputs are archived (§6).

1.4 Outline

Section 2 proves no $m \geq 2$ dicritical exists for any Keller map. Section 3 proves the $m = 1$ two-branch bridge is empty for every degree. Section 5 maps each script to the arithmetic it certifies. Section 6 describes the computational archive. Section 7 gives the AI disclosure.

2 No dicritical of order $m \geq 2$

2.1 The constant-leading-form mechanism

Let E_k ($k \geq 2$) be an exceptional divisor in the resolution, with local coordinates (u', v') in which $E_k = \{v' = 0\}$. For a polynomial $S \in K[x, y]$ of degree d, write the pullback as

$$S = v'^{-\nu} U(u', v') \cdot (\text{monomial in } u'), \tag{1}$$

where $\nu = \nu_{E_k}(S)$ is the order of vanishing and U is a unit (nonzero at the center).

Proposition 2.1 (Constant leading form)**.** *Suppose E_k ($k \geq 2$) is dicritical of order $m \geq 1$ for the pencil. Then after the standard normalization, the leading coefficient satisfies* $\deg_{u'} U(u', 0) = 0$.

Proof sketch. The dicritical condition of order m requires the u'-degree of the pencil restriction to equal m. Tracking the unit U through the blowup sequence: each blowup at a center with $u_0 \neq 0$ multiplies the leading coefficient by u_0^{-2} (a nonzero constant). The u'-dependence cannot increase. A direct computation in the normalized chart shows the u'-degree of $U(u', 0)$ is 0. The verification is in `general_constant_form.py` (§5). □

Corollary 2.2. *No E_k ($k \geq 2$) is dicritical of order $m \geq 2$ for a Keller map.*

Proof. If E_k were dicritical of order $m \geq 2$, Proposition 2.1 gives $\deg U(u', 0) = 0$, but the order-m condition requires $\deg = m \geq 2$, a contradiction. The middle-divisor (E_2) and leaf cases (E_3, E_4) are verified in `middle_e2.py`, `leaf_e3.py`, `leaf_e4.py`. □

2.2 The first exceptional divisor

On E_1, local geometry does permit a solitary $m = 2$ dicritical: the linear conditions for $v_{E_1} \geq -2$ admit a 12-dimensional span at $d = 4$ containing pairs with exact order -2 and exact pencil degree 2 (verified in `e1_solitary.py`, `e1_global.py`).

However, this local survivor is incompatible with the Keller condition:

Theorem 2.3 (E_1 degree obstruction)**.** *Let P, Q lie in the E_1 $m = 2$ open subset (so $v_{E_1} = -2$ and both pencil restrictions have exact degree 2). Then $[P, Q]$ is never a nonzero constant.*

Proof. Every monomial $x^J y^K$ allowed by the order condition $v_{E_1} \geq -2$ satisfies $J+K \geq 2$. Hence:

- P_x has minimum total degree 1 (no constant term);
- P_y has minimum total degree 1;
- similarly Q_x, Q_y have no constant term.

Therefore $[P, Q] = P_x Q_y - P_y Q_x$ has minimum total degree 2. It cannot be a nonzero constant. The computation is in `e1_keller_search.py`. □

Corollary 2.4. *No dicritical of order $m \geq 2$ exists for any Keller map.*

Proof. All E_k ($k \geq 2$) are closed by Corollary 2.2. The remaining case E_1 is closed for Keller maps by Theorem 2.3. □

3 The $m = 1$ bridge is empty

3.1 Setup

Consider the two-branch bridge topology: two distinct centers $a_3 \neq a_4$ (both nonzero), each requiring order $\mathrm{ord}_b(S) \geq 2d - 1$ (the $m = 1$ condition). For $S = \sum p_{JK} x^J y^K$, the pullback at center a is

$$S_a(b) = \sum_{J,K} p_{JK} (a + \tilde{a} b)^{d-J-K} b^{2d-2J-K}, \tag{2}$$

and we require the coefficients of $b^0, \ldots, b^{2d-2}$ to vanish at $a = a_3$ and $a = a_4$.

Theorem 3.1 (Bridge collapse). *For arbitrary distinct nonzero a_3, a_4 and arbitrary $d \geq 2$, the simultaneous conditions* $\mathrm{ord}_b(S) \geq 2d - 1$ *at both centers force $p_{JK} = 0$ whenever $J \geq 1$. The nullspace is exactly* $\mathrm{span}\{1, y\}$.

In particular, every P, Q in the span is a function of y alone, so $[P, Q] \equiv 0$: no Keller pair exists in the $m = 1$ bridge.

3.2 Descending induction on weighted degree

Order monomials by the weighted degree $W = 2J + K$, descending. For fixed W, write the variables as (J_k, K_k), $k = 1, \ldots, n$, ordered by descending J_k. Define $e_k = d - J_k - K_k = d - W + J_k$; the e_k are strictly decreasing (hence distinct).

The equations for weight W come from the pairs (t, i) with $2d - t + i = W$ (where t is the b-power and i the $\tilde{a}$-power), at both centers. The (t, i, a) equation is:

$$\sum_{k=1}^{n} p_{J_k K_k} \binom{e_k}{i} a^{e_k - i} = 0, \qquad a \in \{a_3, a_4\}. \tag{3}$$

Lemma 3.2 (Block full rank). *For distinct $e_1, \ldots, e_n$ and $m \geq n$, the $2m \times n$ matrix with entries $\binom{e_k}{i} a^{e_k - i}$ ($i = 0, \ldots, m - 1$, $a \in \{a_3, a_4\}$) has rank n.*

Proof. Factor column k by $a^{e_k} \neq 0$ and row (i, a) by $a^{-i} \neq 0$; rank is preserved. It suffices to show $\left[\binom{e_k}{i}\right]$ ($i = 0, \ldots, n - 1$) has rank n. Since $\binom{e}{i}$ is a polynomial in e of exact degree i with leading coefficient $1/i!$, elementary row operations transform this matrix to the scaled Vandermonde $[e_k^i / i!]$. As the e_k are distinct,

$$\det\left[\binom{e_k}{i}\right] = \frac{\prod_{j<k}(e_k - e_j)}{\prod_{i=0}^{n-1} i!} \neq 0. \tag{4}$$

The $n = 3$ case is verified symbolically in `m1_block_rank.py`, giving $-(e_1 - e_2)(e_1 - e_3)(e_2 - e_3)/2$. □

Proof of Theorem 3.1. Descend on W from $2d$ to 2. Assume all variables with weight $> W$ are zero. The equations (3) for weight W involve only variables of weight $\geq W$ (higher weights contribute 0 by the inductive hypothesis). By Lemma 3.2, the block has full column rank, forcing $p_{J_k K_k} = 0$ for all k. In particular, every variable with $J \geq 1$ vanishes. The only survivors are $(J, K) = (0, 0)$ and $(0, 1)$, i.e. $\mathrm{span}\{1, y\}$. □

3.3 Computational verification

The induction is verified computationally:

- Symbolic nullspace at general $a_3 \neq a_4$: $d = 2$ gives nullity 2; $d = 3$ gives nullity 2 (`m1_bridge_general.py`).
- Numeric nullspace at $a_3 = 2$, $a_4 = 5$: nullity 2 for $d = 2, \ldots, 8$ (`m1_induction.py`).
- Block-triangular structure exhibited at $d = 6$ (`m1_triangular.py`).
- Vandermonde determinant verified symbolically (`m1_block_rank.py`).

Corollary 3.3. *The $m = 1$ two-branch bridge contains no Keller pair, for every degree $d \geq 2$.*

4 Proof of the main theorem

Proof of Theorem 1.1. Suppose a Keller map admitted two distinct dicritical divisors. Each has some order $m \geq 1$. If either has $m \geq 2$, Corollary 2.4 gives a contradiction. Hence both have $m = 1$. Two $m = 1$ dicriticals form either an adjacent pair or a bridge. The adjacent case is closed by the $d = 3$ computation (`e3_adjacent.py`, `routeB_cancellation.py`); the bridge case is closed for every d by Corollary 3.3. Contradiction. □

5 Script-to-arithmetic correspondence

Each script below was actually executed; outputs are archived in `outputs/`. The table maps every script to the exact paper statement it certifies.

Script	Paper locus	Certifies
`general_constant_form.py`	Prop. 2.1	Leading-unit degree count
`middle_e2.py`	Cor. 2.2	E_2 not dicritical, $m \geq 2$
`leaf_e3.py`	Cor. 2.2	E_3 leaf, $m = 2$ impossible
`leaf_e4.py`	Cor. 2.2	E_4 leaf, $m = 2$ impossible
`e1_solitary.py`	§3.2	E_1 $m = 2$ span is 12-dim at $d = 4$
`e1_global.py`	§3.2	Explicit pair, $v = -2$, deg $= 2$
`e1_keller_search.py`	Thm. 2.3	J has min degree 2, never constant
`m1_bridge_general.py`	Thm. 3.1	Symbolic nullspace, $d = 2, 3$
`m1_induction.py`	§4.3	Nullity 2 for $d = 2, \ldots, 8$
`m1_triangular.py`	§4.2	Block-triangular structure at $d = 6$
`m1_block_rank.py`	Lem. 3.2	Vandermonde determinant $\neq 0$
`at_degree_bound.py`	Cor. 2.2	Finite bridge table, $m \leq 4$
`at_degree_bound_m5.py`	Cor. 2.2	Finite bridge table, $m \leq 5$
`mixed_poles.py`	Cor. 2.2	Mixed $(2, 3)$ bridge
`depth5_global.py`	Cor. 2.2	Depth-5 enumeration
`depth6_enum.py`	Cor. 2.2	Depth-6 enumeration
`routeB_cancellation.py`	Thm. 1.1	Adjacent $m = 1$ at $d = 3$
`e3_adjacent.py`	Thm. 1.1	E_2–E_3 adjacent pair

6 Computational archive

All scripts used in this work are archived in `scripts/` with their outputs in `outputs/`:

- 18 Python scripts (SymPy-based), each with a header stating its epistemic status;
- 18 output files recording exact stdout and exit codes;
- All scripts exited with code 0 on re-execution (October 1, 2026).

Reproducibility

All artifacts are archived in the public repository, directory `branch_two_dicritical/`. Each script runs with `~/miniconda3/envs/physics/bin/python` (Python 3.12, SymPy). The verification script `verify_all.sh` re-runs every script and checks exit codes.

7 Computational and AI-assisted research disclosure

Muse (Meta), the author's personal AI assistant, assisted with computational scripting, manuscript drafting, and proof verification; all script outputs were verified by actual execution. The author (B. D. McCrary) directed the investigation, verified the results, and takes responsibility for the manuscript. The mathematical argument stands on the artifacts above, not on these attributions.

References

[1] Guccione, J. A., Guccione, J. J., Horruitiner, R., Valqui, C. *The Jacobian conjecture for the degree pair* $(72, 108)$. arXiv:2204.14178v1 (2022).

[2] McCrary, B. D., Muse. *The* $(72, 108)$ *Candidate for the Planar Jacobian Conjecture: A Containment Note.* Zenodo, 2026. doi:10.5281/zenodo.22941057.